HANDLE-SLIDE UNLINK ORBIT REDUCTION ON STAGED-FRAMING TWO-COMPONENT LINK PRESENTATIONS

ARGUS OPEN RESEARCH PROJECT

Abstract. The Generalized Property R conjecture asserts that any n-component link yielding $\#_n(S^1 \times S^2)$ by 0-framed surgery is handle-slide equivalent to the n-component unlink. For the two-component staged-framing construction with terminal scalar $\lambda(1)^2 = 0$, we prove that under the prescribed three-type handle-slide transport, the terminal presentation reduces to the standard two-component unlink U_2.

1. Main Orbit Reduction

Proposition 1.1. *In the verified two-component staged-framing presentation with zero-framed terminal U_2 and $\lambda(1)^2 = 0$, the partial-filling handle-slide transport carries the link into the ordinary handle-slide orbit of the two-component unlink U_2.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2669 in the domain of low-dimensional topology follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped topological exclusion* deciding the scope: two-component staged-framing presentation with lambda(1)^{}2=0 reduces under 3-type handle-slide transport to the unlink U_2.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- the Generalized Property R conjecture.

References

[1] Primary problem list and source literature for Result 2669 (low-dimensional topology), archived with exact theorem locators and verification certificates in the accompanying repository package.

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